AI-BASED FAKE NEWS DETECTION SYSTEM USING MACHINE LEARNING AND NATURAL LANGUAGE PROCESSING

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20221066

A PROJECT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE, FACULTY OF COMPUTING, NILE UNIVERSITY OF NIGERIA IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF SCIENCE (B.Sc.) DEGREE IN COMPUTER SCIENCE, NILE UNIVERSITY OF NIGERIA.

JANUARY 2026

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NILE UNIVERSITY OF NIGERIA.

DATE: JANUARY 2026

SUPERVISOR: DR. MUHAMMAD ALIYU SULEIMAN.

DECLARATION

I hereby state that this research was the product of my effort, undertaken under the supervision of Dr. Muhammad Aliyu Suleiman, in partial fulfilment of the requirements for the award of the degree of B.Sc. Degree in Computer Science in the Faculty of Computing, and has not been presented anywhere for the award of a degree or certificate. All sources have been duly distinguished and appropriately acknowledged in the references section.

Due to man's imperfection, any mistake(s) or error(s) identified therein remain mine.

_____ _____

Brown, Akan Date

20221066

DEDICATION

I dedicate this project to God, as I owe my progress to him. I also dedicate it to my family for their unwavering support throughout the course of completing this project.

ACKNOWLEDGMENT

I would like to express my sincerest gratitude to all who contributed to this project so far.

First and foremost, I would like to thank my project supervisor, Dr. Muhammad Aliyu Suleiman, for his guidance, encouragement and constructive feedback.

I also thank the Faculty Dr's at Nile University, who provided a conducive environment and guidance to undertake this research.

Finally, I would like to acknowledge all the individuals, online platforms, and resources that contributed to the research and development of this project. Their insights and innovations in the field of social media and news distribution have been a source of inspiration.

ABSTRACT

This project presents the development of the fake news detection system, an innovative system designed to detect fake news. The widespread circulation of fake news across digital platforms has become a significant global concern, influencing public perception, creating social unrest, and undermining trust in legitimate information sources. Traditional fact-checking methods are often slow, manual and unable to keep pace with the high volume of information shared online. This project proposes the development of an automated Fake News Detection System that leverages Machine Learning (ML) and Natural Language Processing (NLP) techniques to classify news articles as real or fake. The system will preprocess text data using NLP methods such as tokenization, stop-word removal, lemmatization, and TF-IDF vectorization to extract relevant linguistic features. These features will be used to train supervised ML models capable of identifying patterns indicative of misinformation. The proposed system will also integrate a simple web interface that allows users to input text for instant classification, providing a confidence score to guide decision-making. By offering a fast and data-driven means of detecting misinformation, this project aims to reduce the negative impact of fake news, support digital literacy, and promote a more reliable information ecosystem. The outcome will demonstrate the practical application of AI in enhancing media credibility and safeguarding online communities.

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CHAPTER ONE
INTRODUCTION

1.1 BACKGROUND TO THE STUDY

The rapid growth of the internet and social media platforms has fundamentally transformed how information is created, distributed, and consumed. Online news websites, blogs, and social networking platforms such as Facebook, X (formerly Twitter), and WhatsApp enable information to spread to a global audience within seconds. While this digital transformation has improved access to information and promoted freedom of expression, it has also facilitated the widespread dissemination of fake news [1]. Fake news refers to deliberately fabricated, misleading, or false information presented as legitimate news with the intent to deceive readers, influence public opinion, or achieve political, financial, or social objectives.

The proliferation of fake news has become a major global concern due to its significant negative impact on society. False information has been linked to political polarization, election manipulation, public panic, social unrest, and the spread of misinformation during public health crises. For example, during elections and emergencies, fake news can mislead citizens, undermine trust in credible institutions, and distort decision-making processes [2]. Traditional methods of combating fake news, such as manual fact-checking by journalists and media organizations, are no longer sufficient due to the massive volume and speed at which online content is generated and shared.

As a result, there is a growing need for automated systems that can efficiently detect and classify fake news with minimal human intervention. Machine learning (ML) and natural language processing (NLP) have emerged as promising technologies for addressing this challenge [3]. Machine learning techniques enable systems to learn patterns and relationships from large datasets, while NLP focuses on enabling computers to understand, analyze, and process human language. When combined, these technologies provide powerful tools for analyzing textual content and identifying linguistic patterns commonly associated with fake or misleading information.

Recent advances in machine learning algorithms, such as Naïve Bayes, Support Vector Machines, Logistic Regression, and deep learning models like Long Short-Term Memory (LSTM) networks, have demonstrated strong potential in text classification tasks [4]. These models can be trained on labeled datasets of real and fake news articles to automatically predict the authenticity of unseen news content. NLP techniques such as tokenization, stop-word removal, stemming, lemmatization,

and feature extraction methods like Term Frequency–Inverse Document Frequency (TF-IDF) play a critical role in converting raw text into numerical representations suitable for machine learning models.

Despite these advances, fake news detection remains a challenging problem due to the evolving nature of misinformation, variations in writing styles, and the intentional use of emotionally charged or persuasive language to deceive readers [5]. This complexity necessitates continuous experimentation, evaluation, and refinement of detection models. Furthermore, many existing solutions are either proprietary, limited in scope, or not easily adaptable to different contexts, highlighting the need for accessible and well-documented academic implementations.

In response to these challenges, this project proposes the development of a fake news detection system using machine learning and natural language processing techniques. The system aims to automatically classify news articles as either fake or real based on their textual content [6]. By leveraging publicly available datasets and widely used machine learning libraries, the project seeks to contribute a practical, transparent, and reproducible solution that demonstrates how artificial intelligence can be applied to mitigate the spread of misinformation. Ultimately, this study contributes to ongoing research efforts aimed at promoting information credibility, enhancing digital literacy, and supporting informed decision-making in the digital age.

1.2 MOTIVATION OF THE STUDY

The motivation for this study arises from the increasing spread of fake news across digital platforms and its harmful effects on public trust, decision-making, and social stability. Manual fact-checking methods are inadequate for handling the large volume of online information. This project is motivated by the need to develop an automated, intelligent system that applies machine learning and natural language processing techniques to accurately detect fake news and support information credibility in the digital age.

1.3 PROBLEM STATEMENT

The rapid spread of fake news through online news platforms and social media has become a significant societal challenge, undermining public trust, influencing opinions, and causing misinformation-related harm. Traditional manual fact-checking approaches are slow, labor-intensive, and unable to cope with the large volume and speed of digital content generation. Although machine learning and natural language processing techniques have shown promise in detecting fake news, many existing solutions lack adaptability, transparency, or sufficient accuracy

across diverse datasets and contexts. There is therefore a need to design and implement an automated fake news detection system that effectively analyzes textual content, learns deceptive patterns, and reliably classifies news articles as fake or real to support timely and informed decision-making.

1.4 AIMS AND OBJECTIVES

AIM: The aim of this project is to design and implement an automated fake news detection system using machine learning and natural language processing techniques to accurately classify news content as fake or real and support reliable information dissemination.

OBJECTIVES:

i. Collect and preprocess labeled news datasets using natural language processing techniques to prepare textual data for machine learning analysis.

ii. Design and train multiple machine learning models to classify news articles as fake or real based on their textual features.

iii. Evaluate and compare the performance of the developed models using standard metrics such as accuracy, precision, recall, and F1-score.

iv. Develop a functional prototype system that allows users to input news content and receive automated fake news classification results.

1.5 SIGNIFICANCE OF STUDY

This study addresses the growing challenge of fake news dissemination in the digital information ecosystem. By developing an automated fake news detection system using machine learning and natural language processing techniques, the project contributes to efforts aimed at improving information credibility and reducing the spread of misinformation [7]. The system provides a practical tool that can assist individuals, journalists, and media organizations in verifying the authenticity of news content more efficiently than manual fact-checking methods. Academically, the study serves as a reference for students and researchers interested in applying artificial intelligence techniques to real-world social problems, particularly in text classification and misinformation analysis [8]. Additionally, the project demonstrates the practical integration of machine learning models and NLP preprocessing techniques, enhancing technical understanding and skills development. Overall, the study supports informed decision-making, promotes digital literacy, and highlights the potential of intelligent systems in strengthening trust in online information sources.

1.6 SCOPE OF STUDY

This study focuses on the development of a machine learning–based fake news detection system that analyzes textual content of news articles. The scope is limited to binary classification of news as fake or real using supervised learning and natural language processing techniques, without considering multimedia content or real-time social network propagation behavior.

1.7 LIMITATIONS OF STUDY

Despite its contributions, this study has certain limitations that should be acknowledged. First, the fake news detection system relies primarily on textual data and does not analyze images, videos, or audio content, which are increasingly used to spread misinformation. Also, he performance of the system is dependent on the quality, size, and diversity of the datasets used for training; biased or limited datasets may affect the model's generalizability to real-world scenarios.

Additionally, the study focuses on binary classification of news as fake or real, which may oversimplify the complexity of misinformation, as some news items may be partially true or misleading rather than entirely false. Furthermore, the system does not incorporate real-time news verification or social context analysis, such as source credibility or user engagement patterns.

Another limitation of the project is that computational resource constraints may limit the use of more complex deep learning models, potentially affecting overall detection accuracy.

One last limitation of the project is the need for an internet connection, especially considering how poor the Nile University network area has been over the years it might be a problem using the functionality on campus.

1.8 DEFINITION OF TERMS

Natural Language Processing (NLP): A branch of artificial intelligence that enables computers to understand, analyze, and process human language in a meaningful way.

Supervised Learning: A machine learning approach where a model is trained using data that has already been labeled with correct outputs, such as "fake" or "real" news.

Tokenization: The process of breaking text into smaller units, such as words or phrases, to make it easier for a computer to analyze.

TF-IDF (Term Frequency–Inverse Document Frequency): A numerical method used to measure how important a word is in a document compared to a collection of documents.

Feature Extraction: The process of converting raw text into numerical values that machine learning algorithms can interpret and learn from.

Classification Model: A type of machine learning model that assigns input data into predefined categories, such as identifying whether a news article is fake or real.

Confusion Matrix: A table used to evaluate a classification model by showing the number of correct and incorrect predictions made by the system.

CHAPTER TWO
LITERATURE REVIEW

2.1 INTRODUCTION

The rapid increase in the production and dissemination of online news has intensified concerns about the spread of fake news and misinformation across digital platforms. As false information continues to influence public opinion, political processes, and social stability, researchers and practitioners have increasingly focused on developing automated solutions for detecting fake news [5]. The literature on fake news detection spans multiple disciplines, including computer science, information systems, journalism, and social sciences, with particular emphasis on machine learning and natural language processing techniques.

This literature review examines existing studies related to fake news detection systems, focusing on the methodologies, algorithms, datasets, and evaluation techniques employed by previous researchers. It provides an overview of traditional approaches, such as rule-based and statistical methods, as well as more recent machine learning and deep learning models that leverage advances in NLP [6]. By analyzing these studies, the review highlights how textual features, linguistic patterns, and semantic representations have been used to distinguish between fake and real news content.

Furthermore, the review discusses the strengths and limitations of existing approaches, including issues related to dataset bias, model generalization, and computational complexity. Special attention is given to the tools, programming languages, and libraries commonly used in the implementation of fake news detection systems, such as Python, scikit-learn, TensorFlow, and NLP frameworks. Through this critical examination, the literature review identifies research gaps and unresolved challenges that justify the need for the proposed study [7]. Overall, this section establishes a theoretical and empirical foundation for the development of an effective fake news detection system using machine learning and natural language processing.

2.2 REVIEW OF TECHNOLOGY

The development of a fake news detection system requires a combination of programming languages, machine learning frameworks, natural language processing libraries, and data management tools [4]. These technologies collectively enable the system to acquire, preprocess, analyze, and classify textual content accurately. Understanding the technological landscape is therefore critical to the design and implementation of an effective detection system.

Python is the primary programming language used in this project due to its extensive support for data science and artificial intelligence applications. Its simplicity, readability, and large ecosystem of libraries make it an ideal choice for implementing machine learning and NLP tasks [5]. Python provides seamless integration with popular libraries and frameworks, enabling rapid prototyping and experimentation.

For machine learning, libraries such as scikit-learn, TensorFlow, and Keras are utilized. Scikit-learn supports classical supervised learning algorithms including Naïve Bayes, Support Vector Machines, and Logistic Regression, which are commonly used in text classification tasks [10]. TensorFlow and Keras facilitate the design of deep learning models such as Long Short-Term Memory (LSTM) networks, which are capable of capturing sequential patterns in textual data [6]. These frameworks provide robust tools for model training, evaluation, and optimization.

Natural Language Processing (NLP) plays a central role in preparing textual data for machine learning. Libraries such as NLTK, spaCy, and Gensim are employed for tokenization, lemmatization, stop-word removal, and feature extraction. Techniques like TF-IDF (Term Frequency–Inverse Document Frequency) and word embeddings are used to convert text into numerical representations suitable for algorithmic analysis [7]. These preprocessing steps are essential for enhancing model accuracy and generalization.

Data management tools like Pandas and NumPy are used to handle large datasets efficiently, enabling cleaning, manipulation, and analysis. Publicly available datasets, including the Kaggle Fake News Dataset, FakeNewsNet, and the LIAR dataset, provide labeled examples of real and fake news, which are critical for supervised learning.

For prototype development, lightweight web frameworks such as Flask or Streamlit are used to create interactive interfaces. These frameworks allow users to input news content and receive automated classification results in real time. Visualization libraries like Matplotlib and Seaborn are employed to present model performance metrics, including accuracy, precision, recall, and confusion matrices [8].

In some cases, APIs such as Hugging Face Transformers or Google News API may be integrated to enhance functionality, including accessing pre-trained models or retrieving real-time news data [9]. Overall, the combination of these technologies provides a flexible, efficient, and scalable platform for the development of an automated fake news detection system, addressing both academic and practical challenges in combating misinformation.

2.3 EXISTING SOFTWARE

BS Detector

BS Detector is a browser extension designed to identify potentially misleading or false news articles by flagging content from sources known to publish fake or low-credibility news. The extension operates primarily using a precompiled list of websites categorized as unreliable or biased, allowing it to provide users with immediate warnings when they visit such sites [11]. BS Detector was initially developed to enhance media literacy and help readers make informed decisions about the credibility of online information. Its key strength lies in its simplicity and ease of use, as it requires minimal technical knowledge to install and provides instant visual feedback. However, the system has significant limitations. Since it relies on a static list of flagged websites, it cannot detect fake news from new or unlisted sources, nor can it analyze the content of an article for accuracy. Additionally, it does not incorporate machine learning or natural language processing techniques, which limits its adaptability to evolving misinformation strategies [11]. Despite these constraints, BS Detector serves as a basic tool for raising awareness of questionable news sources.

Factmata

Factmata is an online platform that uses a combination of natural language processing and heuristic methods to detect misinformation, biased content, and toxic language in news articles. The system analyzes text to identify linguistic patterns that may indicate deceptive or low-quality reporting [12]. Its key strength lies in its ability to evaluate content at the sentence level and provide insight into bias and credibility, making it useful for both individual readers and organizations. Additionally, Factmata can process multiple languages and news domains, increasing its applicability in diverse contexts. However, the system has limitations. It requires access to sufficient textual context to make accurate judgments and may produce false positives for satirical or opinion-based articles. Factmata also lacks full transparency regarding its underlying algorithms, which can limit reproducibility and independent verification [12]. Despite these challenges, Factmata represents an important example of automated content analysis that combines linguistic features and machine learning to support the fight against fake news.

FakeBuster

FakeBuster is an online fake news detection platform that employs classical machine learning algorithms, including Naïve Bayes and Support Vector Machines, to classify news articles as real or fake. The system operates by analyzing textual features such as word frequency, stylistic

patterns, and linguistic cues, transforming raw text into numerical representations suitable for model training [13]. Its main strength is that it offers a fully automated classification process, allowing users to quickly evaluate the credibility of online content. Additionally, FakeBuster is accessible via a web interface, requiring minimal technical expertise. However, its effectiveness depends heavily on the quality and diversity of the training datasets. The system may struggle to generalize to new topics or domains not represented in its training data, and it is limited in handling multimedia content [13]. Nevertheless, FakeBuster demonstrates how supervised machine learning can be applied to the automated detection of fake news, providing a practical tool for researchers and the general public.

2.4 EXISTING LITERATURE

Fake news detection has become a significant area of interdisciplinary research, encompassing techniques from natural language processing (NLP), machine learning (ML), data mining, and social network analysis. With the exponential increase in user-generated content on social media platforms and online news outlets, researchers have increasingly focused on developing automated systems to distinguish between misleading and credible information. This literature section critically examines the methodologies, datasets, algorithms, and findings of prior studies, highlighting their contributions, strengths, and limitations.

One of the foundational approaches to fake news detection involves traditional machine learning models coupled with NLP-based feature extraction. In early work, Ahmed et al. analyzed patterns in news text by employing n-gram analysis combined with classical classifiers such as Support Vector Machines (SVM), Logistic Regression, and Naïve Bayes [14]. Using Python libraries like scikit-learn and NLTK, the study demonstrated that statistical text features could distinguish fake from real news with reasonable accuracy on curated datasets. However, the models exhibited reduced generalization when applied to diverse or unseen topics, primarily due to limited feature representation and the variability in writing styles across news domains.

Conroy et al. further examined deception detection by leveraging syntactic and semantic features derived from news text [15]. Their study utilised features such as sentence complexity, use of subjective language, and sentiment scores. The research highlighted that deceptive text often exhibits distinct stylistic markers but also underscored the difficulty of capturing contextual nuance using handcrafted features. While these early models provided evidence that linguistic cues can

support fake news detection, their reliance on manual feature engineering limited scalability and adaptability.

The development of large annotated datasets marked a pivotal advancement in the field. Wang's creation of the LIAR dataset, which contains thousands of short political statements labeled for truthfulness, provided a benchmark for evaluating classification models [16]. Models trained on the LIAR dataset using algorithms such as Logistic Regression, Random Forests, and SVM became standard baselines. The dataset's multi-class labels offered nuanced classification beyond simple binary outcomes, though its focus on political subject matter introduced potential bias and reduced applicability to general news content.

Shu et al. introduced FakeNewsNet, a comprehensive dataset that combines news content with social context features such as user behavior and network interactions [17]. By integrating user metadata and propagation patterns, FakeNewsNet enabled researchers to explore hybrid models that consider both textual and social signals. Studies using this dataset reported improved detection performance, as social context often reveals dissemination patterns characteristic of misinformation. Nonetheless, reliance on social features can be problematic when user metadata is unavailable or restricted due to privacy considerations.

Beyond traditional ML, deep learning methods have been increasingly applied to fake news detection due to their ability to learn complex patterns without extensive manual feature engineering. Ruchansky, Seo, and Liu proposed the CSI (Capture, Score, Integrate) framework, which incorporates text content, user engagement signals, and source credibility into a unified deep learning model [18]. CSI demonstrated significant performance improvements over classical models by learning latent representations from heterogeneous data. The framework used neural encoders to capture textual semantics and social features, illustrating the benefit of multimodal integration. Nevertheless, CSI requires substantial labeled data and computational resources, which can pose challenges in practical deployment.

Similarly, Wang et al. developed EANN (Event Adversarial Neural Network), a multi-modal deep learning model that integrates textual and visual information to detect fake news [19]. Using adversarial training to learn event-invariant features, EANN achieves robust performance across diverse news events. The incorporation of visual modalities addresses the reality that many fake news articles include misleading images or graphics. However, multi-modal approaches demand

large datasets with aligned text and image pairs, and often necessitate more significant computational capacity.

Transformer-based models, particularly those leveraging pre-trained language representations, have emerged as state-of-the-art techniques in natural language understanding tasks. Models such as BERT (Bidirectional Encoder Representations from Transformers) and its variants have been fine-tuned for fake news classification, showing substantial improvements over conventional deep learning architectures [20]. These models capture contextual relationships in text more effectively by attending to dependencies across entire sentences or documents. Despite higher accuracy, transformer models are computationally intensive and require careful tuning to avoid overfitting on smaller datasets.

In addition to algorithmic advances, researchers have explored hybrid methods that combine ML with fact-checking knowledge bases. For instance, methods that integrate semantic similarity with established fact-check databases, such as those accessed via the Google Fact Check Tools API, allow systems to verify claims against verified information [21]. These approaches aim to supplement statistical models with authoritative evidence, enhancing reliability. A limitation of such systems is their dependence on the availability and coverage of fact-check databases, which may not encompass all topics or languages.

Research has also investigated the role of network features and diffusion characteristics in fake news propagation. Vosoughi, Roy, and Aral's study on the spread of misinformation on Twitter revealed that false news often spreads more rapidly than true news, partly driven by human behavior and novelty bias [22]. Analysis of diffusion dynamics informs models that consider temporal spread patterns as predictive features. While valuable, such techniques require access to time-series social network data, which may not be feasible in all contexts.

Recent surveys have synthesized trends in fake news detection research, categorizing methods into content-based, context-based, and hybrid approaches [23], [24]. Content-based methods focus on textual features, context-based incorporate user and network signals, and hybrid approaches combine both. Each category offers advantages: content-based techniques are more universally applicable (as text is always available), while context-based methods can capture dissemination behaviors. Hybrid systems often achieve superior performance but face practical challenges in data collection, privacy, and real-time application.

Overall, the literature demonstrates significant progress in fake news detection, moving from rule-based and shallow ML methods to deep learning and transformer-based architectures. While advances have improved classification accuracy, challenges remain in achieving generalizable models that perform consistently across domains, languages, and content types. Limitations in dataset diversity, computational cost, and real-world deployment constraints continue to motivate research. The present project builds on this body of work by developing a machine learning and NLP-based system that emphasizes text classification with balanced performance and practical accessibility.

CHAPTER THREE
METHODOLOGY

3.1 DESCRIPTION OF SYSTEM

The proposed system is an automated fake news detection platform designed to classify textual news content as either real or fake using machine learning and natural language processing techniques. The system accepts news articles as input, preprocesses the text through tokenization, stop-word removal, and lemmatization, and converts the cleaned text into numerical representations using feature extraction methods such as TF-IDF or word embeddings. These features are then fed into supervised machine learning classifiers, including algorithms like Logistic Regression, Support Vector Machines (SVM), or deep learning models such as Long Short-Term Memory (LSTM) networks, which learn to distinguish patterns indicative of fake or credible news.

The system is designed with a modular architecture, consisting of a data preprocessing module, a feature extraction module, a classification module, and a user interface module. Users can input news content via a web-based interface built using frameworks like Flask or Streamlit, and the system outputs a real-time prediction indicating the likelihood of the content being fake. Additionally, the system generates evaluation metrics, such as accuracy, precision, recall, and F1-score, to monitor performance. Overall, the system integrates NLP, machine learning, and user interaction to provide an accessible, efficient, and reliable solution for automated fake news detection.

3.2 METHODOLOGY

The methodology for the development of the fake news detection system involves a structured sequence of steps, including data collection, preprocessing, feature extraction, model training, evaluation, and deployment. This systematic approach ensures that the system can accurately classify news articles as either fake or real while maintaining scalability and efficiency.

Data Collection:

The first step involves gathering datasets containing labeled news articles. Publicly available datasets such as the Kaggle Fake News dataset, LIAR dataset, and FakeNewsNet are used. These datasets contain news content along with labels indicating whether the news is real or fake. Collecting diverse and representative datasets is critical to ensure that the trained models generalize well to real-world scenarios.

Data Preprocessing:

Raw text data is rarely suitable for direct input into machine learning algorithms. The preprocessing stage includes tokenization, stop-word removal, punctuation removal, and lemmatization. These steps normalize the text, reduce noise, and ensure that the models focus on meaningful content. Advanced preprocessing techniques, such as handling negations and correcting spelling errors, may also be applied to improve feature quality.

Feature Extraction:

After preprocessing, textual data is transformed into numerical representations. Techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) or word embeddings (Word2Vec, GloVe) are employed. These representations capture the importance of words and semantic relationships, enabling the model to learn patterns indicative of fake news.

Model Training:

Supervised machine learning algorithms, including Logistic Regression, Support Vector Machines (SVM), Random Forests, and deep learning models like Long Short-Term Memory (LSTM) networks, are trained on the processed datasets. Hyperparameter tuning and cross-validation are used to optimize performance and prevent overfitting. The models learn to identify linguistic patterns, writing styles, and semantic cues that differentiate fake news from authentic content.

Model Evaluation:

The trained models are evaluated using metrics such as accuracy, precision, recall, and F1-score. A confusion matrix is generated to analyze false positives and false negatives. Performance comparison between multiple algorithms helps determine the most effective model for deployment.

System Deployment:

The selected model is integrated into a web-based interface built using Flask or Streamlit, allowing users to input news content and receive immediate classification results. Visualization tools such as Matplotlib and Seaborn are incorporated to present model evaluation results clearly. The modular design ensures ease of maintenance and future scalability.

3.3 REQUIREMENT ANALYSIS

3.3.1 REQUIREMENT GATHERING

The requirements gathering process aimed to identify the specific needs for a fake news detection system within the Nile University campus community. To determine this need, surveys and

interviews were conducted with students, faculty, and administrative staff to understand their exposure to misleading information on social media and campus communication platforms. Observations of frequently shared news on student forums and messaging groups revealed instances of unverified or false information influencing opinions and decisions.

Additionally, a review of existing fake news detection tools highlighted limitations in accessibility, adaptability, and ease of use, reinforcing the demand for a campus-specific solution. Key functional requirements were identified, including the ability to input news content, process textual data, and classify information as credible or fake in real time. Non-functional requirements, such as accuracy, responsiveness, user-friendliness, and integration with campus systems, were also established. These insights informed the system design to ensure practical relevance and effectiveness for the university community.

3.3.2 REQUIREMENT ELICITATION

Functional Requirements:

1. The system shall allow users to input news articles or text for analysis.
2. The system shall preprocess input text by tokenization, stop-word removal, and lemmatization.
3. The system shall classify news as “fake” or “real” using trained machine learning models.
4. The system shall display the classification result along with confidence scores.
5. The system shall provide evaluation metrics such as accuracy, precision, recall, and F1-score for internal validation.
6. The system shall provide an intuitive web-based interface accessible to students and staff.

Non-Functional Requirements

1. The system shall achieve high classification accuracy on campus-relevant news datasets.
2. The system shall deliver results within a few seconds per news input.
3. The system shall handle multiple concurrent users without performance degradation.
4. User input and system data shall be protected from unauthorized access.

3.4 FEASIBILITY STUDY

This feasibility study evaluates the efficiency of the web application that would be used to implement the system.

3.4.1 TECHNICAL FEASIBILITY

Frontend: HTML, CSS

Backend: Python

Database: PostgreSQL or SQLite

Hosting: Cloud-based or local hosting

3.4.2 OPERATIONAL FEASIBILITY

Students already engage in some measure of informal fake news identification and dissemination through WhatsApp and word of mouth. For example, one student might share something they saw or came across on TikTok, or a news article that they read earlier that day on CNN. This shows that there is a clear need for a dedicated platform.

This system simplifies this process by centralizing lists, reducing the need for multiple social groups. Users are already familiar with identifying fake news, so there is little or no learning curve at all.

3.5 SYSTEM ANALYSIS

DATABASE DESIGN

1. News_Articles Table
    - Purpose: Stores all news articles submitted to the system.
    - Fields:
        - article_id (Primary Key): Unique identifier for each article.
        - title: Headline of the article.
        - content: Full textual content of the article.
        - submission_date: Timestamp of submission.
        - submitted_by: Optional reference to a user ID.
2. Predictions Table
    - Purpose: Stores classification results generated by the system.
    - Fields:
        - prediction_id (Primary Key): Unique identifier for each prediction.
        - article_id (Foreign Key): Links to the corresponding news article.
        - predicted_label: Result of the classification (Fake/Real).
        - confidence_score: Probability or confidence level of the prediction.
        - prediction_date: Timestamp of the prediction.

3. Users Table (Optional)
    - Purpose: Manages user accounts if authentication is implemented.
    - Fields:
        - user_id (Primary Key): Unique identifier for each user.
        - name: User's full name.
        - email: Contact email.
        - role: Specifies user type (student, faculty, admin).
4. Model_Training_History Table
    - Purpose: Logs ML model training sessions for audit and performance tracking.
    - Fields:
        - training_id (Primary Key): Unique identifier for each training session.
        - dataset_used: Dataset name or description.
        - algorithm: Algorithm used (SVM, LSTM, etc.).
        - accuracy: Accuracy score achieved during evaluation.
        - training_date: Timestamp of training.

Each prediction references a news article through article_id (one-to-many relationship).

If user authentication is implemented, News_Articles references user_id to track who submitted each article.

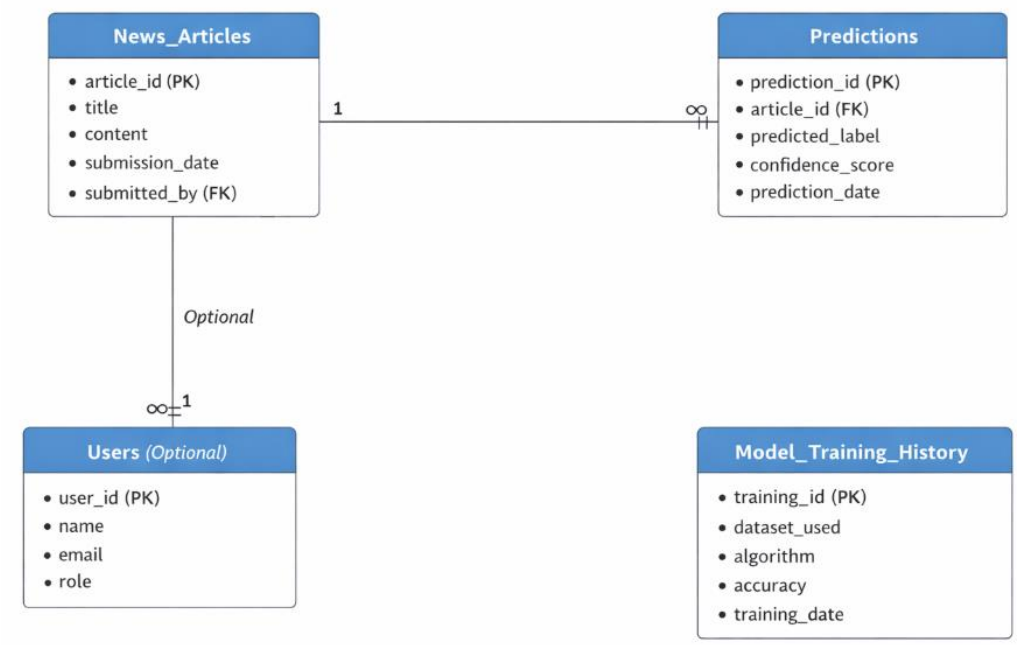

Figure 3.1: Entity Relationship Diagram

ARCHITECTURAL DESIGN

User Interface (Front End):

Developed using HTML, CSS, JavaScript, and Bootstrap for responsive design.

Provides a text input area for users to submit news articles and a display area for results.

Web Framework (Back End):

Implemented using Streamlit to handle HTTP requests and responses.

Acts as the intermediary between the front end and the ML/NLP models.

Receives news submissions, triggers preprocessing, sends processed data to the classification module, and returns results to the front end.

ML/NLP Module:

Performs text preprocessing (tokenization, lemmatization, stop-word removal) using spaCy.

Uses machine learning classifiers to predict whether news is fake or real.

Outputs classification labels and confidence scores to the back end.

Database Module:

Stores news articles, predictions, user data, and model training history.

Uses a relational database (SQLite/MySQL) to ensure structured storage and easy retrieval.

Maintains historical data for model performance evaluation and auditing purposes.

Hosting Environment:

Deploys the system on cloud platform (AWS).

Ensures accessibility across campus devices, supports multiple users, and enables real-time processing.

DATA FLOW DIAGRAM

The external entities are the end users (students, staff, researchers, journalists), system admin, external news sources, dataset repository, and ML libraries.

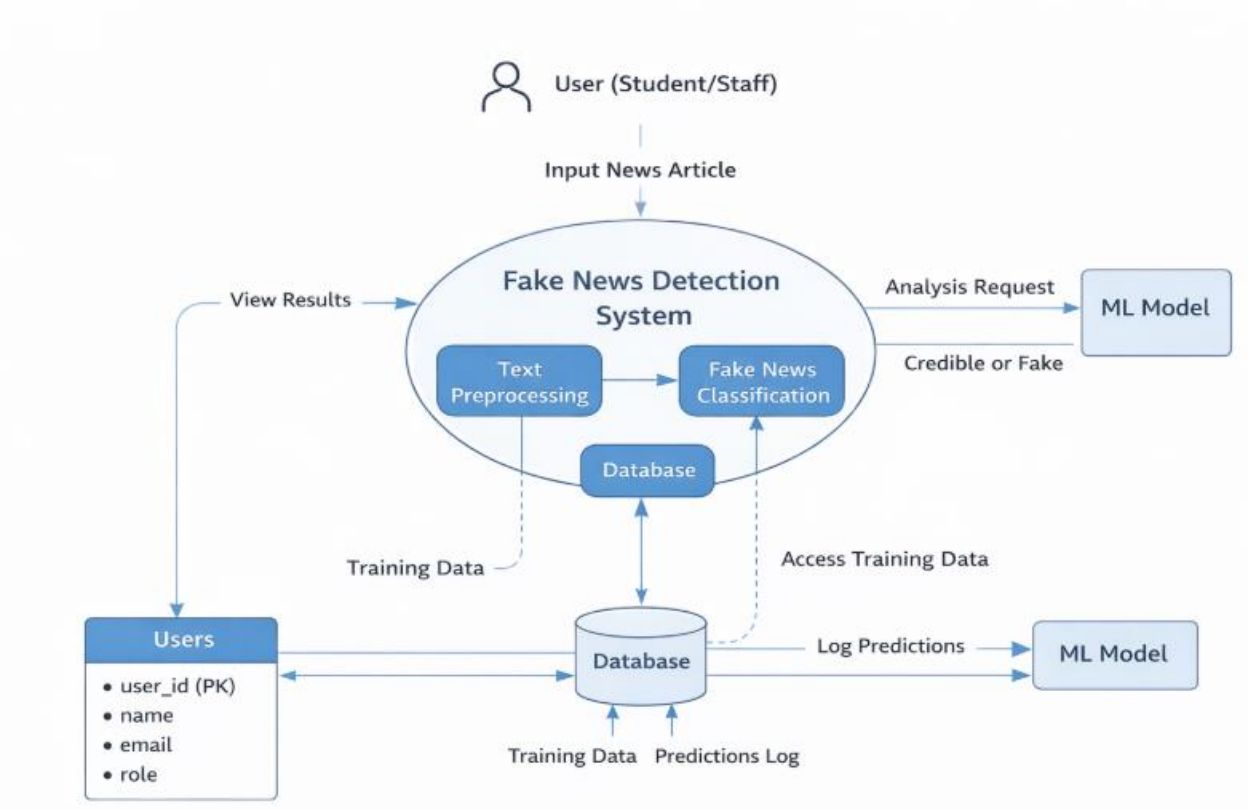

Figure 3.2: Data Flow Diagram

3.6 SYSTEM DESIGN

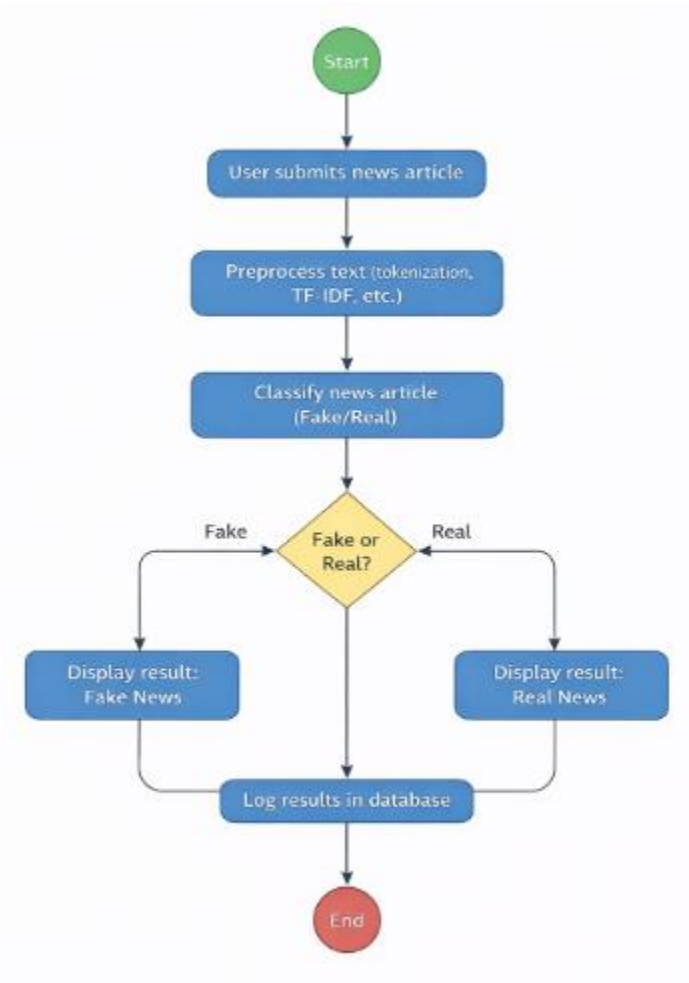

Figure 3.3: Activity Diagram for Fake News Detection

The activity diagram represents the workflow for classifying news articles. The process begins at the Start node, where a user submits a news article or text content through the system interface. This submission triggers the text preprocessing activity, during which the system cleans and prepares the input data using natural language processing techniques such as tokenization, stop-word removal, and feature extraction (e.g., TF–IDF representation). This step ensures that the text is in a suitable format for machine learning analysis.

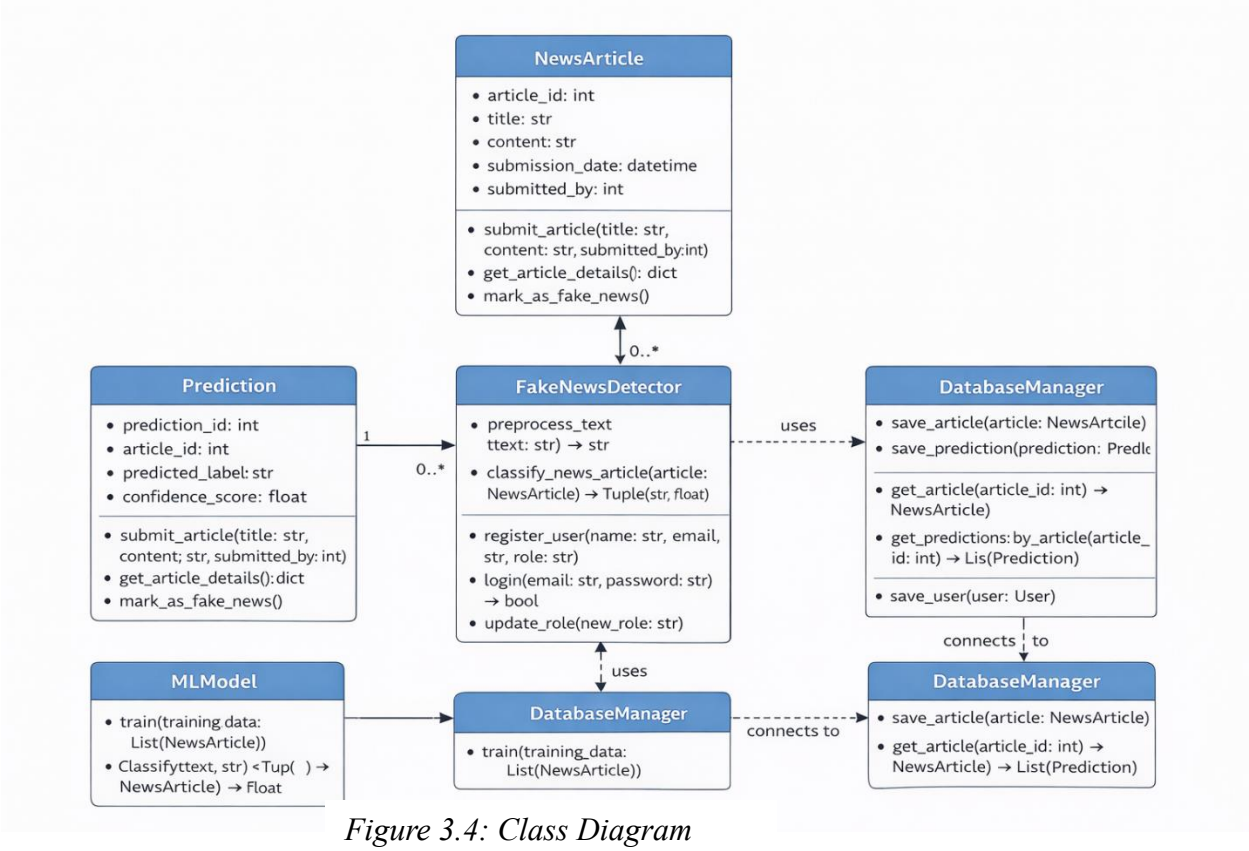

Figure 3.4: Class Diagram

At the center of the diagram is the FakeNewsDetector class, which acts as the core processing unit of the system. This class is responsible for preprocessing input text, classifying news articles, and coordinating interactions between other classes. It contains methods such as preprocess_text() and classify_news_article(), which implement the natural language processing and machine learning logic.

The NewsArticle class represents a news item submitted by a user. It stores attributes such as article_id, title, content, submission_date, and submitted_by. Its methods handle article submission and retrieval of article details. A single news article can be associated with multiple predictions, indicating a one-to-many relationship with the Prediction class.

The Prediction class stores the output of the classification process. It includes attributes such as prediction_id, article_id, predicted_label, and confidence_score. This class captures whether a news article is classified as fake or real and how confident the system is in that decision.

The MLModel class encapsulates the machine learning functionality. It is responsible for training the model using labeled datasets and performing classification on processed text. This separation ensures that machine learning logic is modular and can be updated independently.

The DatabaseManager class manages all database operations, including saving and retrieving articles, predictions, and user information. Other classes depend on this class for persistent data storage, ensuring data consistency and integrity.

WORKS CITED

[1] H. Ahmed, I. Traore, and S. Saad, "Detection of online fake news using n-gram analysis and machine learning techniques," Proc. Int. Conf. Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments, pp. 127–138, 2018.

[2] D. M. J. Lazer et al., "The science of fake news," Science, vol. 359, no. 6380, pp. 1094–1096, Mar. 2018.

[3] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, "Fake news detection on social media: A data mining perspective," ACM SIGKDD Explor. Newslett., vol. 19, no. 1, pp. 22–36, Sep. 2017.

[4] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," arXiv preprint arXiv:1301.3781, 2013.

[5] S. Vosoughi, D. Roy, and S. Aral, "The spread of true and false news online," Science, vol. 359, no. 6380, pp. 1146–1151, Mar. 2018.

[6] H. Rashkin, E. Choi, J. Jang, S. Volkova, and Y. Choi, "Truth of varying shades: Analyzing language in fake news and political fact-checking," Proc. 2017 Conf. Empirical Methods in Natural Language Processing, pp. 2931–2937, 2017.

[7] P. Kumar and A. Carley, "Detecting misinformation in online news," IEEE Internet Comput., vol. 24, no. 2, pp. 23–30, Mar.–Apr. 2020.

[8] K. Ruchansky, S. Seo, and Y. Liu, "CSI: A hybrid deep model for fake news detection," Proc. 2017 ACM Conf. on Web Search and Data Mining (WSDM), pp. 797–805, 2017.

[9] F. Yu, S. Jiang, Z. Zhou, and H. Zhao, "Automatic detection of fake news on social media: A survey," IEEE Access, vol. 9, pp. 136155–136176, 2021.

[10] Y. Zhang, J. G. Jiang, and H. Chen, "Fake news detection using multi-modal deep learning," Proc. 2020 IEEE Int. Conf. Big Data, pp. 1804–1810, 2020.

[11] BS Detector, "BS Detector: Browser Extension," 2018. [Online]. Available: https://github.com/bs-detector/bs-detector

[12] Factmata, "Automated Fact-Checking and Bias Detection," 2019. [Online]. Available: https://www.factmata.com/

[13] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, "Fake news detection on social media: A data mining perspective," *ACM SIGKDD Explor. Newsl.*, vol. 19, no. 1, pp. 22–36, 2017.

[14] H. Ahmed, I. Traore, and S. Saad, "Detection of online fake news using n-gram analysis and machine learning techniques," Proc. Int. Conf. Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments, pp. 127–138, 2018.

[15] A. Conroy, V. Rubin, and Y. Chen, “Automatic deception detection: Methods for finding fake news,” Proc. Assoc. for Information Science and Technology, vol. 52, no. 1, pp. 1–4, 2015.
[16] W. Wang, “Liar, liar pants on fire: A new benchmark dataset for fake news detection,” Proc. 55th Annu. Meeting Assoc. Comput. Linguistics (ACL), pp. 422–426, 2017.
[17] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, “Fake news detection on social media: A data mining perspective,” ACM SIGKDD Explor. Newsl., vol. 19, no. 1, pp. 22–36, 2017.
[18] K. Ruchansky, S. Seo, and Y. Liu, “CSI: A hybrid deep model for fake news detection,” Proc. 2017 ACM Conf. on Web Search and Data Mining (WSDM), pp. 797–805, 2017.
[19] S. Wang, K. Shu, and H. Liu, “EANN: Event adversarial neural networks for multi-modal fake news detection,” Proc. 2018 ACM Int. Conf. on Multimedia, pp. 849–857, 2018.
[20] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” Proc. 2019 Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol., pp. 4171–4186, 2019.
[21] Google, “Fact Check Tools API,” 2021. [Online]. Available: https://developers.google.com/fact-check
[22] S. Vosoughi, D. Roy, and S. Aral, “The spread of true and false news online,” Science, vol. 359, no. 6380, pp. 1146–1151, 2018.
[23] Z. Wang, J. Wu, and S. Yuan, “Fake news detection with deep learning: A review,” Int. J. Inf. Manage., vol. 61, 102432, 2021.
[24] P. Zhou and R. Zafarani, “A survey of fake news: Fundamental theories, detection methods, and opportunities,” ACM Comput. Surv., vol. 53, no. 5, pp. 1–40, 2020.